

Reese Dunne

Stanford, CA | radunne@stanford.edu | [linkedin.com/in/reesedunne](https://www.linkedin.com/in/reesedunne)

EDUCATION

M.S. & Ph.D. Candidate, Mechanical Engineering, Stanford University | GPA: 3.93/4.00 Expected June 2029

- *Fellowships & Awards*: NSF Graduate Research Fellowship, Stanford Graduate Fellowship, Tau Beta Pi Fellowship
- *Selected Courses*: Deep Learning, Programming Abstractions (C++/data structures), Prob. & Statistics, Linear Algebra, FEA
- **Research Output**: 5 publications (2 first-author) | 7 international conference presentations | [Google Scholar](#)

B.S. Mechanical Engineering, Mississippi State University | GPA: 4.00/4.00 May 2023

- *Selected National Awards*: Rhodes Scholarship Finalist, Knight-Hennessy Finalist, Goldwater Scholar, Astronaut Scholar

TECHNICAL SKILLS

- **Languages & Tools**: Python | MATLAB | Bash/Shell | Fortran | Linux | Stata | Jupyter | Abaqus | C++ | R (familiar)
- **Machine Learning**: TensorFlow | scikit-learn | Deep Learning | NumPy | PyTorch (familiar)
- **Imaging & Data Analysis**: Statistical Analysis | Mixed-Effects Modeling | MRI | Image Registration | Segmentation | FSL

RESEARCH EXPERIENCE

Longitudinal MRI of Repetitive Head Impacts in Collegiate Athletes, Doctoral Researcher Oct. 2025–Present

Dr. Michael Zeineh, Department of Radiology, Stanford University

- Lead a longitudinal neuroimaging analysis of 74 collegiate athletes across 262 MRI examinations, using source-separated QSM and diffusion MRI biomarkers to investigate brain microstructural changes associated with repetitive head impacts over 4 years
- Develop linear mixed-effects models across 4 susceptibility and 8 diffusion MRI metrics to quantify longitudinal trajectories in football vs. volleyball athletes, revealing sport-dependent differences in white matter; presented orally at ISMRM 2026

Quantitative MRI Biomarkers of Alzheimer’s Disease (AD), Doctoral Researcher Jan. 2024–Present

Dr. Michael Zeineh, Department of Radiology, Stanford University

- Develop Python, MATLAB, and Bash pipelines to process and analyze 7T MRI data from 20 healthy/MCI/AD participants, identifying significant increases in hippocampal R2* with greater disease severity, consistent with increased iron
- Analyze 3T PET-MR and source-separated QSM from 66 participants across 5 clinical/biomarker groups, identifying susceptibility changes across 4 memory- and olfactory-related regions and distinguishing iron- vs. myelin-related abnormalities

Mechanical Characterization of Plant-Based and Animal-Based Meat, Doctoral Researcher Sept. 2023–July 2025

Dr. Ellen Kuhl, Department of Mechanical Engineering, Stanford University

- Led a study of 8 meat products using double-compression texture profile analysis and oscillatory rheology to quantify 10 mechanical properties and establish quantitative targets for improving plant-based meat texture; presented orally at ESB 2025
- Conducted 157 tension, compression, and shear tests for a study applying Constitutive Artificial Neural Networks (CANNs) to discover product-specific 3D constitutive models, enabling the first 3D mechanical characterization of 8 plant/animal meats

Computational Modeling of Biodegradable Magnesium Implants, Undergraduate Researcher Aug. 2020–Dec. 2024

Dr. Matthew Priddy, Department of Mechanical Engineering, Mississippi State University

- Built a multiscale computational model combining finite element analysis (12,760-element Abaqus/UMAT framework) and density functional theory to predict magnesium implant degradation over 120 days, showing hydroxyapatite coating extended implant lifespan by ~40 days compared to uncoated magnesium

Diffusion MRI of Exercise-Related Skeletal Muscle Adaptation, Research Intern June 2021–May 2022

Dr. Garry Gold, Department of Radiology, Stanford University

- Created Python-based 3D visualizations and analyzed 25 longitudinal DTI scans using fiber tractography, quantifying changes across 3 quadriceps muscles, including a ~12% increase in rectus femoris fiber length after 3 months of resistance training

Transcranial Photoacoustic Image Simulation, Research Intern May 2020–July 2021

Dr. Muyinatu Bell, Department of Electrical and Computer Engineering, Johns Hopkins University

- Conducted 3D photoacoustic k-Wave simulations comparing compressional vs. elastic wave models across 3 acoustic windows, showing equivalent target detectability with 78% less computation time and 63% less memory, informing pre-surgical planning

SELECTED PUBLICATIONS

- **Dunne RA et al.** “Texture profile analysis and rheology of plant-based and animal meat.” *Food Res. Int.*, 2025. **First author**
- **Dunne RA et al.** “Finite element and density functional theory modeling effectively predict pitting degradation of hydroxyapatite-coated pure magnesium.” *J. Biomed. Mater. Res. B*, 2024. **First author**
- St. Pierre SR et al. “The mechanical and sensory signature of plant-based and animal meat.” *npj Sci. Food*, 2024. **Co-author**
- Graham MT, **Dunne RA**, Bell MAL. “Comparison of compressional and elastic wave simulations for patient-specific planning prior to transcranial photoacoustic-guided neurosurgery.” *J. Biomed. Opt.*, 2021. **Second author**

LEADERSHIP & ACTIVITIES

- NCAA Division I Track & Field / Cross Country Athlete, Mississippi State University (2018–2023)
- Founder, *Nonnie’s Notes* (2016–2023) – led 50+ volunteers serving assisted-living facilities; incorporated as 501(c)(3) in 2022